

Uzbekistan OS API Specification Version 1.0

1. Purpose

Defines the public and internal APIs for the Uzbekistan OS MVP, including architecture, authentication, request/response formats, streaming, error handling, and lifecycle management.

2. API Principles

REST-first, JSON-based, stateless, versioned, typed, predictable, self-documenting, and backwards compatible.

3. Base URL

Versioned API under /api/v1/ with future compatibility for /api/v2 and beyond.

4. Authentication

Guest sessions, JWT authentication for users, role-based access for administrators, and bearer-token authorization.

5. Standard Response Format

Consistent success and error envelopes with data, metadata, and standardized error codes.

6. Authentication Endpoints

Register, login, refresh token, and logout endpoints.

7. User Endpoints

Profile retrieval, profile updates, and user preference management.

8. Conversation Endpoints

Create, retrieve, list, and soft-delete conversations.

9. Messaging & Chat

Streaming chat endpoint, message history, conversation messaging, and citation-aware AI responses.

10. Knowledge APIs

Search, document retrieval, structured content access, and official source references.

11. Workflow APIs

Workflow listing, retrieval, execution, progress tracking, and checklist generation.

12. Feedback API

Capture answer ratings and user feedback for evaluation and continuous improvement.

13. Internal AI APIs

Query understanding, retrieval, answer generation, summarization, checklist generation, embeddings, and validation services.

14. Crawler & Ingestion APIs

Queue crawl jobs, monitor ingestion status, validate structured knowledge, and publish reviewed content.

15. Administration APIs

User management, review queues, document publication, rejection, and moderation.

16. Pagination & Filtering

Cursor-based pagination with standardized filtering and sorting conventions.

17. Streaming

Server-Sent Events (SSE) supporting start, chunk, citation, workflow, done, and error events.

18. Security

HTTPS, JWT validation, rate limiting, schema validation, sanitization, CSRF protections where applicable, audit logging, and least-privilege access.

19. OpenAPI Strategy

Contract-first API design using OpenAPI 3.1 with generated documentation, SDKs, and contract tests.

20. Definition of Done

Every endpoint documented with schemas, authentication, validation, streaming behavior, versioning, testing strategy, and implementation guidance.